BU7852 Group Project, 2026-27

> ***DRAFT, pending Eoin's review. Not yet confirmed with Neil Dunne, not yet posted to students.***

"Is UK business better or worse off after Brexit?"

Worth 50% of the module. Groups of four. Report and a recorded presentation. Your Claude log is required evidence (Green on the module's AI policy).

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1. The question, and why you can't answer it as written

"Better off" is not a number, and "business" is not one thing. A sector can grow and still carry more risk. A firm can survive and still be weaker than it looks. Your first job is to turn the question into one you can actually answer with data, state what it leaves out, and say who you are answering it for (an auditor assessing going-concern risk, a lender, an investor, a minister).

Brexit brought a friend. The UK left the single market on 1 January 2021, in the middle of a pandemic that hit every sector, and government support schemes (furlough, Bounce Back Loans, the temporary suspension of winding-up petitions) changed the normal relationship between a weak economy and company failure. Almost everything that moved in sector output or insolvencies in 2020-22 has at least two candidate causes, and for part of that period a third: policy support actively holding back failures that would otherwise have happened. A good report says plainly what it can and cannot separate. It does not pretend.

> ***An honest "the evidence points this way, this far, and no further" can earn a First. A confident verdict the analysis cannot support will not.***

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2. The dataset pack (everyone gets the same data)

Real, public, UK government data, frozen for the term and posted to Blackboard.

File	What it is	Rows	Source
`uk_sector_output.csv`	UK GDP (output approach), quarterly, chained-volume index for 18 industry sections (manufacturing, construction, retail, finance, hospitality, and so on) plus the total economy, 2010-Q1 to 2026-Q2, rebased so **2016-Q2, the referendum quarter, = 100** for every sector. This file has a genuine pre- and post-referendum baseline.	1,254	ONS, GDP output approach, low-level aggregates
`uk_insolvencies.csv`	UK company insolvencies, the same 18 sections, broken out by nation (**England & Wales**, **Scotland**, and a **Great Britain** total): annual 2016-2022, then quarterly (summed from the monthly releases) 2023-Q1 to 2026-Q2. No type breakdown (compulsory liquidation, administration, and so on) at sector level, that split only exists at the national total. No sector-	1,134	The Insolvency Service, Company Insolvency Statistics

File	What it is	Rows	Source
	level data before 2016, so this file has no real "before," use it for the post-2016 risk picture. Great Britain, not the full UK (Northern Ireland has no industry-level table).		

Rules on data

- Your core analysis must use the pack. That is what gets marked.
- You may bring in extra data to support a point. If you do: cite it, include the file with your submission, and say why the pack wasn't enough.
- You may cite published estimates of the Brexit effect on UK business, and Claude knows several. Quoting someone else's number is not analysis. Marks come from what *you* found in *this* data, and from how you compare it with theirs.

Things in the pack that will catch you out if you don't look (we'll meet the ideas in class):

- Insolvencies fell sharply in 2020, not because businesses were healthier, but because furlough, Bounce Back Loans and a suspension of winding-up petitions were propping firms up, then rose well above their pre-pandemic level once that support ended. The rebound is not "Brexit catching up," and it is not "nothing to do with Brexit" either. Decide what you can and cannot claim.
- An output index tells you a sector's size changed. It does not tell you whether the firms in it are financially healthier or weaker. Pairing the two files, not reading either alone, is the point of requirement 3.
- `uk_insolvencies.csv` has no sector-level data before 2016, so it cannot answer a before/after question by itself. Run your before/after test on `uk_sector_output.csv`, which genuinely spans 2010 to 2026, and use the insolvency file for the post-2016 risk picture in requirement 5.
- The insolvency file gives you a total count per sector, not a type breakdown (compulsory liquidation, administration, and so on). That split exists only at the national level, not crossed with sector. Don't claim a sector-level split the data doesn't have.
- `uk_insolvencies.csv` has a `nation` column with three values: England & Wales, Scotland, and Great Britain (the first two added together). If you group by sector and period without filtering `nation`, you will sum all three and triple-count. Filter to `Great Britain` unless you specifically want the England & Wales vs Scotland comparison.

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3. What your analysis must include

1. **A defined question.** One or two sentences: what "better or worse off" means in your report, measured by which columns, over which years, for which audience (auditor, lender, investor, policymaker). Justify the choice, including what it misses.
2. **A confidence interval and a hypothesis test, before and after.** Using the methods from Sessions 1-6, on `uk_sector_output.csv`, estimate a confidence interval for your chosen sector's output and test whether it changed from before the referendum (and/or the end of the transition period) to after. **You must state your comparison sector or sectors, justify the choice, and show what happens to your answer if you choose a different one.** This is the heart of the project, the same move as the undergrad module's comparator-country requirement, run on BU7852's own tools.
3. **One more method from the course, your choice:**
   - *A chi-square test*, on `uk_insolvencies.csv`: group the 18 sectors however you define and justify (goods-producing vs services, trade-exposed vs domestic, or your own split), and test whether that

grouping is independent of whether a sector's insolvencies rose or fell between two years of your choosing.
    - *Logistic regression*: using each sector's pre-2016 output (level and growth, from `uk_sector_output.csv`) as predictors, how well can you predict whether its insolvencies went on to rise? Eighteen sectors is a small sample for this: say what that does to your confidence in the result.
4. **The confound, handled explicitly.** What did you do about Covid and the support schemes that followed it, and what can't you separate?
5. **Who carries the risk.** At least one paragraph on which sectors (or, if you use the England & Wales vs Scotland split, which nation) carry more of the risk your analysis finds, written as if briefing an audit partner building a going-concern or sector-risk watchlist. This is your bridge to BU7856.
6. **A recommendation.** If an audit partner, a lender, or a minister asked "so what, and where should I be looking?", what would you tell them, and how confident are you?

Python in Colab with Claude, as in class. Submit your notebook alongside the report.

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4. Deliverables and timing

Dates below are proposed against the 26-27 teaching calendar (Tuesdays from 29 Sept, 8 sessions, in-class test in Session 6) and need confirming once the full session plan is locked.

What	Detail	Proposed
Groups of four	Self-selected. Names in by Session 3. After that, whoever is left gets placed.	Session 3
Question check-in (not graded)	One paragraph: your defined question, your comparison sector(s), your second method. Feedback within a week.	Session 5
Project clinic	Bring your draft analysis.	Session 8
Recorded presentation	Max 8 minutes, every member appears and speaks, findings not a methods tour. No live Q&A (per the module's assessment format).	~2 weeks after Session 8
Report + notebook + Claude log	Report max **2,000 words** (tables, charts, appendix excluded). Group cover sheet with the AI declaration.	Same date as the presentation

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5. How it is marked (50 marks)

Component	Marks	What earns the marks
Framing the question	7	A precise, measurable question; honest about what the choice leaves out; a stated audience.
Analysis	13	Sound confidence interval and hypothesis test; one further course method; comparison sector justified and **tested** by changing it.
Interpretation and the confound	10	Numbers read correctly; claims sized to the evidence; Covid and the support schemes dealt with explicitly; correlation not sold as cause.
Who carries the risk	6	Sector, firm-size or regional distribution named; written for an audit/risk audience; limits of the data and method stated.

Component	Marks	What earns the marks
Data visualisation	4	Charts that answer the question at a glance, honestly scaled, correctly labelled.
Recorded presentation	5	Clear story, every member contributes, fits in 8 minutes.
Claude log and reflection	5	See section 6.

Every component is marked in four bands: **Fail / Pass / Merit / First**.

What separates a First from a Pass: a Pass report reads well and restates what is generally known about Brexit's effect on UK business. A First report shows the reader something they could only learn from *this* analysis: how much the answer moves with the comparison sector, where the Covid problem bites and where it doesn't, which sectors or firm types the average hides. It is careful about what it cannot claim.

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6. Using Claude

Claude is required, as in class. It is also the easiest way to hand in a fluent, generic report that earns a Pass. Claude has read a great deal about Brexit's economic effects, and if you ask it this question it will answer with other people's published estimates. Use it to *do* your analysis, not to *be* your analysis.

Submit:

- **The full Claude conversation log(s)** for the project (export, don't curate).
- **A reflection, max 500 words**, with the group's name on it: one place Claude helped you think better, and **one place Claude was wrong, overconfident, or quietly steered you, and how you caught it.** A reflection with no example of catching Claude out cannot reach Merit.

The log is marked on the quality of your questions and how critically you treated the answers, the same standard as in class.

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7. Group work

- Groups of four. If a group falls to three, tell Eoin early. Scope stays the same.
- Include a short **contribution statement** in the appendix: who did what, signed by all four.
- If a group has a serious contribution problem, raise it before the deadline, not after.

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8. Keeping it civil

This is a politically charged question and people in the room will have strong views, some personal. The project asks *what the evidence supports*, not *whether Brexit was right*. You can reach any conclusion the analysis supports. You will lose marks for claims the analysis doesn't support, whichever side they favour.